

Homework 2

ASEN 5335 Aerospace Environment

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September 12, 2026

Problem 1: Spaceballs

(i.)

We are able to use the provided MSISatmosphere1000.m profile to calculate the an approximate total mass of the atmosphere. The MSISatmosphere1000 function takes inputs of altitudes and outputs a structure, one of which is mass density which corresponds to that altitude. Using MATLAB we are able to have a vector of altitudes that range to 1000 km, the maximum model input of the MSISatmosphere1000 function. We can then obtain a vector of mass densities ρ . We use the following,

$$m = \rho V$$

$$m = \int \rho dV$$

And for a sphere,

$$m = \int \rho * 4\pi r^2 dr$$

$$\therefore m = 4\pi \int \rho r^2 dr$$

With ρ being a vector and r also being a vector of radius of this atmospheric shell (altitude including radius of the Earth $R_E = 6371$ km), we can integrate within MATLAB to obtain a

$$\boxed{m = 5.297 * 10^{18} \text{ kg}}.$$

(ii.)

Assuming a volume of $V = 10 * 10^3 * 10 * 10^3 * 100 * 10^3 = 1 * 10^{13} \text{ m}^3$, a constant temperature $T = 300 \text{ K}$, molar mass $M = 0.029 \text{ kg/mol}$, and universal gas constant $R = 8.314 \text{ J/(molK)}$. We can use the ideal gas law,

$$PV = \frac{m}{M}$$

Rewrite to solve for pressure P ,

$$P = \frac{m}{MV} RT$$

$$\boxed{P = 4.556 * 10^{10} \text{ Pa}}$$

Problem 2: Atmospheric Scale Height

To calculate the scale height H of air mixture assuming N_2/O_2 mixture at sea level, we use the equation.

$$H = \frac{kT}{mg}$$

Where we assume a constant $T = 300$ K from the problem, and $k = 1.38 * 10^{-23}$ J/K, the Boltzmann constant, $g = 9.81$ m/s², the gravitational constant. We then seek the molecular mass of N_2/O_2 mixture. This can be obtained from the table provided in the HW. We find the atomic mass of the air mixture provided with the % by volume

$$amu_{air} = 28.0134 (.781) + 31.9988 (.209) = 28.566 \text{ g/mol}$$

$$m_{air} = amu_{air} * \frac{1 \text{ kg}}{1000 \text{ g} * 6.02 * 10^{23} \text{ g/mol}}$$

$$m_{air} = 4.745 * 10^{-26} \text{ kg}$$

This process of finding molecular weight and converting it can be repeated for atomic Oxygen, helium, and hydrogen.

$$amu_{oxygen} = (31.9988 \text{ g/mol})/2 = 15.9994 \text{ g/mol}$$

$$amu_{helium} = 4.0026 \text{ g/mol}$$

$$amu_{hydrogen} = 2.01594 \text{ g/mol}$$

$$m_{oxygen} = 2.6577 * 10^{-26} \text{ kg}$$

$$m_{helium} = 6.6488 * 10^{-27} \text{ kg}$$

$$m_{hydrogen} = 3.3487 * 10^{-27} \text{ kg}$$

The scale height equation can be used with each respective molecular mass of the species in question.

$$H_{air} = 8.8938 * 10^3 \text{ m}$$

$$H_{oxygen} = 1.588 * 10^4 \text{ m}$$

$$H_{helium} = 6.347 * 10^4 \text{ m}$$

$$H_{hydrogen} = 1.2602 * 10^5 \text{ m}$$

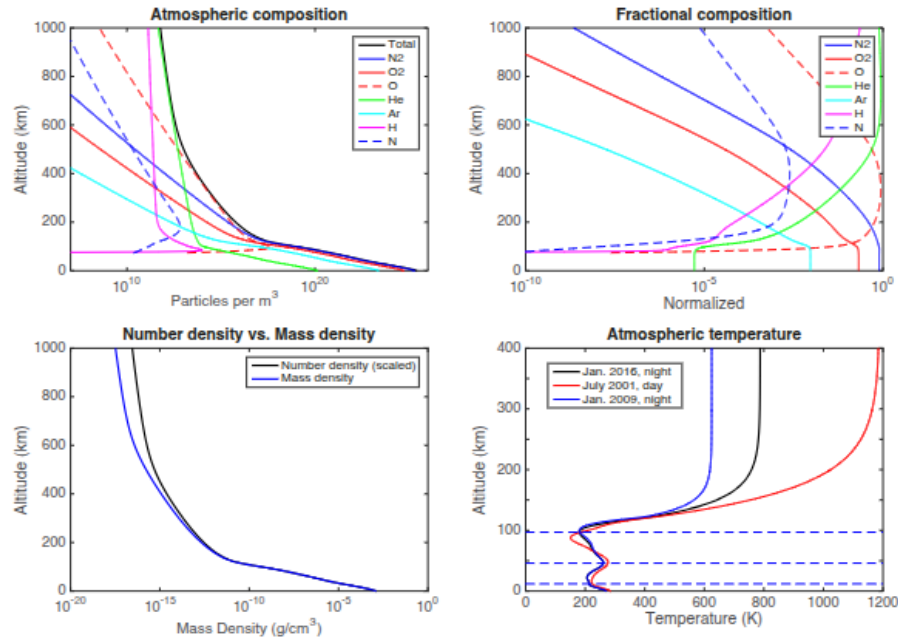


Figure 1: Atmospheric Profile

By comparing these scale heights to those in the Figure 1 provided from the HW, we observe that

TODO: We seek to also calculate scale height from the MSIS profiles. From the MSIS profiles there will be different temperatures instead of the fixed T used earlier. Using altitudes of 150,180,450,500,600,1000km.

Problem 3: Atmosphere Variation

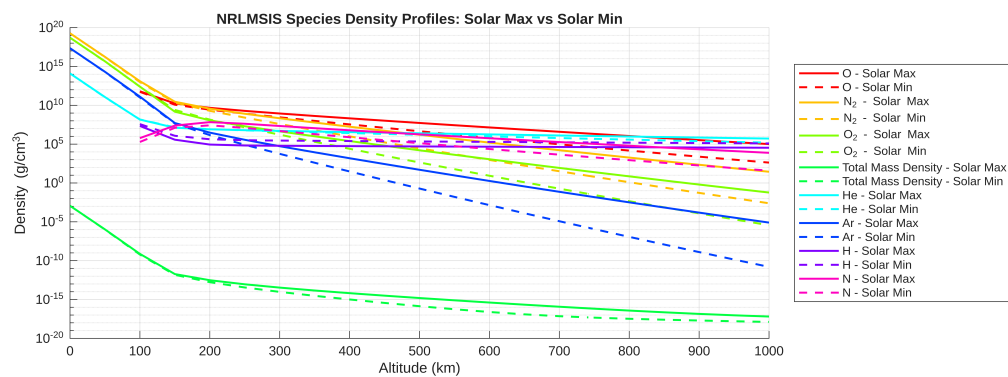


Figure 2: NRLMSIS Species Density Profiles Solar Max vs Solar Min

Problem 4: Oxygen Impact

Problem 5: Orbit Decay

Problem 6: Atmosphere Uncertainty and Neutral Winds

Problem 7: Air Breathing Thruster

Problem 8: Missions